

JOHN (JACK) PETERS

ROBOTICS | CONTROLS | MECHATRONICS

Nashville, TN | jackpeters06798@gmail.com | [LinkedIn](#) | [Website](#)

PROFESSIONAL SUMMARY

Robotics engineer and Ph.D. candidate with 8+ years of industry and research experience spanning robot design, modeling and control, mechatronics, electrical system design, and clinical integration. Develops robotic systems from mathematical modeling and trajectory optimization through mechatronic design, controls software, and experimental validation. Expertise includes continuum robotics, minimally invasive surgical robotics, image-guided intervention, design optimization, and data-driven control.

TECHNICAL SKILLS

Robotics & Controls: Robot Kinematics & Dynamics, Trajectory Optimization, System Identification, Optimal, Nonlinear, Adaptive, & Data-Driven Control

Software: MATLAB/Simulink, C++, Python, ROS2, Git, Ubuntu, 3D Slicer

Hardware & Mechatronics: SolidWorks, 3D Printing & Rapid Prototyping, Pneumatic Actuation, Electromechanical Design, Sensor Integration, PCB Design, System Integration & Testing

EXPERIENCE

Vanderbilt University - [MEDLab](#) / [C* Controls Lab](#)

Nashville, TN

Graduate Research Assistant | Ph.D. Candidate, Mechanical Engineering

Aug 2021 - Present

- Designed and experimentally validated a compact 6-DOF, MRI-compatible robotic system for minimally invasive neurosurgery, achieving 83.5% mean cannulation of target tissue in MRI-guided experiments versus clinical averages of 50-60%.
- Developed a constrained optimization framework that jointly optimizes concentric tube robot design and actuation to maximize conformal ablation coverage of complex bilateral brain tumors; achieved 91% mean coverage across a 31-patient imaging cohort.
- Developed data-driven modeling and control methods for soft and continuum robots, using experimentally learned models for real-time task-space control; demonstrated millimeter-scale tracking on both pneumatic and tendon-driven platforms.
- Designed compact pneumatic stepper actuators for in-bore robotic intervention, reducing size by 73% while increasing actuation force/torque by 3x and preserving MRI image quality.
- Extended the neurosurgical robot from curved needle delivery to conformal therapy by integrating a directional laser probe and real-time MR thermometry monitoring; achieved 84.2% mean hippocampal ablation coverage versus the reported 56% clinical average in tissue-mimicking phantoms.
- Advanced the robotic system toward clinical translation by validating an end-to-end neurosurgical workflow in cadavers, integrating fluoroscopic skull-base access, MRI-guided robotic deployment, and laser ablation.

ASML – Photolithography Systems

Wilton, CT

Electrical Design Engineer II

Jul 2018 - Jul 2021

- Designed electrical infrastructure for advanced semiconductor lithography systems from concept through prototype integration, test, and introduction to volume manufacturing.
- Developed schematics and wiring architectures spanning custom PCBs, AC/DC power distribution, data acquisition/control consoles, and stage/motor control assemblies.
- Integrated and troubleshoot complex electromechanical systems to component level in laboratory and cleanroom environments; authored design specifications, test procedures, and engineering reports.
- Supported FPGA/HDL development, firmware updates, and configuration management using Intel Quartus and Git; produced designs with Mentor Graphics DX/Xpedition.

Medtronic – Hugo™ Robot Assisted Surgery System

North Haven, CT

Engineering Associate

2016 - 2018

- Supported mechatronic development of the Hugo™ robotic-assisted surgery (RAS) system, including circuit design, prototype assembly, and component-level troubleshooting.
- Designed and executed hardware and system-level test procedures and documented engineering results to support design verification and development decisions.

SELECTED ROBOTICS PROJECT

Open-Source Concentric Tube Robot Simulation & Teleoperation

[GitHub: JackyP1937/ctr-sim](#)

Independent Project | Python, ROS2, Git

Summer 2026

- Developed an open-source simulation and control framework for concentric tube robots in ROS2/Python, implementing Cosserat-rod Newtonian mechanics, tube-torsion modeling, and kinematic control.
- Implemented real-time task-space teleoperation using an Xbox controller with resolved-rate inverse kinematics and online, warm-started Jacobian updates.
- Organized the simulator as a modular ROS2 package separating robot mechanics, kinematics, control, and visualization, including automated testing and Git-based version control.

EDUCATION

Vanderbilt University - 2026

Ph.D., Mechanical Engineering - Expected Dec 2026 | M.S., Mechanical Engineering
Co-advisors: Eric J. Barth, PhD & Robert J. Webster III, PhD

Nashville, TN

GPA: 3.96 / 4.00

Selected Graduate Coursework: Robotic Manipulators, Optimal Control, Machine Learning for Dynamical Systems, Nonlinear Control, Adaptive Control

University of Connecticut - 2018

B.S.E., Double Major: Electrical Engineering & Biomedical Engineering | Summa Cum Laude

Storrs, CT

GPA: 3.97 / 4.00

TEACHING & LEADERSHIP

Graduate Teaching Assistant, Vanderbilt University - Multiple appointments in Applied Robotics and Engineering Senior Design; Led laboratory instruction, programming industrial serial manipulators (Denso, Mecademic, Mitsubishi) using C++ and ROS2.

SELECTED PUBLICATIONS & INTELLECTUAL PROPERTY

- **Peters, J. E.**, et al. "An MRI-Guided Robotic System to Improve Hippocampal Access for Epilepsy Interventions." Annals of Biomedical Engineering. In review. [arXiv:2609.13572](https://arxiv.org/abs/2609.13572)
- **Peters, J. E.**, et al. "Conforming to Curvature: Improving Hippocampal Ablation Coverage with Robotic, Curvilinear, Directional MRgLITT." IEEE Transactions on Biomedical Engineering. In preparation.
- Grillo, A.M., **Peters, J.E.**, et al. "MR-Guided Robotic Transforaminal Laser Interstitial Thermal Therapy for Epilepsy: Initial In-Scanner Cadaver Experience." IEEE Transactions on Biomedical Engineering. In review.
- Kumar, N.S., **Peters, J.E.**, et al. "Constrained Lagrangian-Based, Nonholonomic Modeling and Validation of a Pneumatically Actuated Wheeled Soft Robot." Journal of Mechanisms and Robotics. [Published](#). 2026.
- Kumar, N.S., **Peters, J.E.**, Beckers, T., Barth, E.J. "Data-driven Dynamic Modeling and Control of Pneumatic Soft Actuators with Stability Analysis." ASME Letters in Dynamic Systems and Control. Accepted. 2026.
- Kumar, N.S., **Peters, J.E.**, Barth, E.J. "Real-Time Data-Driven Task-Space Control of Tendon-Drive Continuum Robots." ASME/IEEE Transactions on Mechatronics. In preparation.
- **Peters, J.E.**, Esser, D.S., Grillo, A.M., Barth, E.J., Webster III, R.J., "Robotic surgical system, apparatus, and method." U.S. Patent Application Publication US20260000870A1, Jan. 2026.

HONORS & AWARDS

NSF Graduate Research Fellowship Program (GRFP) Honorable Mention, 2023 | University Graduate Fellowship (UGF) | Graduate Leadership Institute Professional Development and Training Grant